To Jungle or to Carry? Estimating Causal Role Impact in Professional League of Legends

Johannes Heinrich[1], Michael Kamp[1,2], and Osman Mian[1,2]

[1] Lamarr Institute, TU Dortmund, Germany
[2] Institute for AI in Medicine, University Medicine Essen
{firstname.lastname}@tu-dortmund.de

Abstract. Quantifying player contribution in E-sports remains difficult because role-specific performance statistics are strongly confounded by team strength, drafting, side selection, and broader match context. We study this problem in professional *League of Legends* (LoL) by analysing the causal influence of the Attack Damage Carry (Adc) and Jungle (Jng) roles on competitive outcomes. Using professional match data from Oracle's Elixir spanning 2014–2025, we transform player-level records into team-level observations with role-specific features and augment them with OpenSkill-based measures of competitive strength. We then estimate latent Adc and Jng performance constructs from gameplay indicators and define an explicit graphical causal model to estimate their effects on match victory and rating gain through backdoor adjustment. Finally, we evaluate robustness through sensitivity analysis and additional causal validity checks to distinguish genuine strategic influence from purely correlational patterns in professional play. The results do not support the claim that Adc is inherently weak in professional play; measured Adc performance shows a larger direct effect on match victory than Jng performance. More broadly, the study illustrates how causal analysis can complement predictive sports analytics by separating role impact from contextual correlations.

Keywords: Causal inference · E-Sports analytics · League of Legends · Latent variable modelling

1 Introduction

E-sports offer a rare favourable setting for sports analytics: professional matches produce dense, structured, and role-specific data at a scale rarely available in traditional sports. This makes competitive games a natural testing ground for machine learning methods that seek to measure player impact, evaluate strategy, and explain match outcomes. Yet most E-sports analyses remain associational. A statistic could correlate with winning either because it reflects genuine impact (strong teams distribute resources differently), or because other factors (such as draft quality, side selection) confound the outcome. As in traditional sports analytics, understanding who actually drives competitive success therefore requires moving beyond associations and towards causal estimation.

We study this problem in professional *League of Legends* (LoL), a leading role-based E-sports. In a standard professional match, two teams of five players compete, with each player assigned a specialised role. We focus on the Attack Damage Carry (Adc) and Jungle (Jng) roles because they sit at opposite ends of a long-running community debate. Players often view the former as resource-dependent and comparatively weak, while they see the Jungler as a high-agency role due to map mobility, lane pressure, and objective control. These claims, however, rarely receive causal treatment and instead rely on gameplay intuition or ranked-play experience rather than professional match-level causal estimates. These competing views naturally motivate a causal question over role impact: given team-level match statistics together with role-specific performance indicators for the Adc and Jng positions, can we perform a causal analysis of their *actual* influence on overall match outcomes? Although existing machine learning approaches in E-sports analysis mainly identify statistical associations between gameplay variables and winning [12, 1], these relationships do not necessarily reflect genuine strategic influence—individual performance emerges from coordinated team play shaped by drafting, macro strategy, and team strength, making purely correlational analysis difficult to interpret reliably.

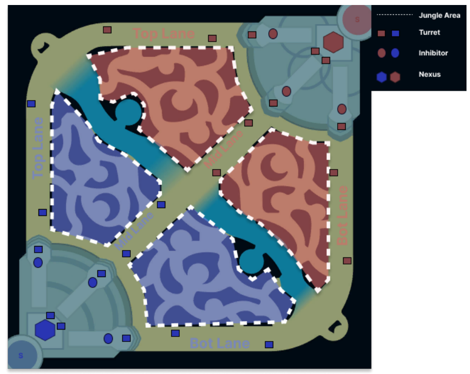

Fig. 1. Illustration, based on "Summoners Rift" map from League of Legends. Teams progress through lanes and defensive structures to reach and destroy the opposing Nexus (denoted by hexagon), the ultimate win condition of a match. See supplementary material for details. Original map is intellectual property of Riot Games[16].

A causal perspective aims to distinguish true impact from confounding competitive factors and to better understand how alternative Adc or Jng performances could influence match outcomes under comparable conditions. In particular, instead of asking the associational question *whether Adc and Jng performance is linked with match outcomes?*, our analysis aims to answer the causal question *how changing Adc or Jng performance would alter match outcomes if everything else in the game were held constant*? In doing so, we move beyond descriptive importance scores towards a more principled understanding of strategic influence. Such an approach supports explanations that enables more reliable interpretation of role-specific contributions to winning, and helps more robustly predict the result of a strategy change within games.

In this work, we estimate the causal influence of Adc and Jng performance under an explicit graphical causal framework. Using professional match data from Oracle's Elixir, comprising 70,000 professional LoL matches collected across 11 years [13], we augment the dataset with rating-based variables derived from the OpenSkill framework [10] and perform a role-specific causal analysis of competitive impact. Our contributions are threefold:

1. We construct an augmented professional LoL dataset and formulate an explicit graphical causal model that captures the dependency structure between team strength, role-specific performance, and match outcomes[1].
2. Using the dataset and our proposed causal graphical model, we estimate the causal effects of ADC and JNG performance on match outcomes and rating gains using established causal inference methods, including backdoor adjustment and outcome modelling.
3. We evaluate the robustness of our causal estimates through sensitivity analysis and statistical robustness tests grounded in the causal inference literature.

2 Related Work

E-sports has become an active area of research across sports analytics, game studies, psychology, and data science. Reitman et al. [15] provide a broad literature review of E-sports research and document the rapid growth of academic work on organized competitive gaming. Within E-sports, LoL is particularly suitable for quantitative analysis because professional matches provide structured match logs, role assignments, and detailed player statistics. Novak et al. [12] analyse performance indicators at the 2018 LoL World Championship, demonstrating the use of statistical performance analysis in professional E-sports. Bahrololloomi et al. [1] propose role-aware player-performance metrics for predicting LoL match outcomes. These works are closely related to our setting, but they are primarily descriptive or predictive rather than causal.

A central challenge in LoL analytics is that player contribution is role-dependent. The same raw statistic can have different meanings for an ADC, Jungler, Support, or Mid Laner. Sharpe et al. [18] investigate competitive LoL performance indicators through a systematic review and focus group study, identifying performance variables across several domains. This supports our view that role performance should be treated as a multidimensional construct rather an isolated statistic. In that regards, rating systems such as TrueSkill [8] and OpenSkill [10] provide important metrics for team-based games, that can be used as outcome for role performance assessment.

In contrast to prior LoL analytics, which mostly studies performance indicators or match prediction, we ask a causal question: how does measured ADC or JNG performance affect professional match outcomes under explicit adjustment assumptions? This allows us to analyse the common claim that ADC is inherently weak relative to JNG without relying only on correlational comparisons.

[1] The dataset and corresponding supplementary material is available at www.github.com/kampmichael/LoLRolePerformance_Causal/

3 Problem Setup and Preliminaries

Game Setup. LoL is a 5v5 multiplayer online battle arena game in which two teams compete to destroy the opposing Nexus. Each player occupies one of five specialised roles: Top, Jungle (Jng), Middle, Attack Damage Carry (Adc), and Support. These roles differ in their strategic responsibilities, resource allocation, and contribution to map control, team fighting, and objective pressure. We provide a general overview of match structure and objectives in Appendix A and focus on the Adc and Jng roles because they influence the game through markedly different mechanisms. The Adc typically serves as the primary sustained damage source in mid- and late-game engagements and relies heavily on gold income, item scaling, and positioning. In contrast, the Jng exerts influence through map mobility, lane pressure, neutral-objective control, and tempo. As a result, identical gameplay statistics can reflect fundamentally different forms of contribution depending on the role that generates them. This role dependence complicates direct performance comparisons and motivates a causal approach. Given team-level match statistics and role-specific indicators, we seek to estimate the causal effect of measured Adc and Jng performance on competitive outcomes under comparable competitive conditions.

Dataset. The dataset consists of structured match data from professional League of Legends matches, spanning from 2014 to 2025. We obtained match data from Oracle's Elixir [13] and merged all competitive seasons into a single dataset. We then removed incomplete matches, excluded variables with substantial missingness or limited relevance, and retained only observations with complete values for the remaining variables. We restructured the data to a team-level format, representing each match with one observation per team. During this process, we aggregated team-level statistics while retaining role-specific indicators relevant to the analysis. We stored role-specific variables using role prefixes (e.g., `adc_kills` and `jng_damagetochampions`), whereas unprefixed variables such as `kills` denote team-level aggregates. Finally, we augmented the dataset with variables derived from the rating model. The resulting dataset contains approximately 70,000 team-level observations and includes five categories of variables: identifiers (e.g., `gameid`), match context (e.g., `gamelength`), team performance metrics (e.g., `kills`, `deaths`), role-specific metrics (e.g., `adc_earnedgold`, `jng_kills`), and rating-based variables (e.g., `win_prob`, `rating_after`).

Causal Analysis. Given team-level match statistics together with role-specific indicators for the Adc and Jng positions, we seek to estimate the causal effect of measured role performance on competitive outcomes. Unlike traditional E-sports analytics, which focuses on observed associations between gameplay variables and winning, causal inference asks a counterfactual question: how would match outcomes change if Adc or Jng performance were different while all other relevant competitive conditions remained comparable? We formalise these assumptions using a causal graph, where role performance acts as the treatment, match victory or rating gain acts as the outcome, and variables such as team strength, side selection, drafting, and match context act as potential confounders. Using the *backdoor criterion*, we identify a sufficient adjustment set that blocks non-causal paths between treatment and outcome and estimate Average Treatment Effects (ATEs). Intuitively, ATE measures the expected change in outcome under alternative levels of role performance after accounting for observed confounding factors. This framework aims to isolate the direct contribution of ADC/JNG performance from broader competitive influences. We refer interested readers to Pearl [14] for a comprehensive introduction to causal inference.

4 Causal Role Impact Estimation

Our objective is to answer the following questions for measured performances of ADC and JNG.

1. *How would win probability change under different levels of ADC performance?*
2. *How would win probability change under different levels of JNG performance?*
3. *How would rating gain change under different levels of ADC performance?*
4. *How would rating gain change under different levels of JNG performance?*

Our causal model is represented by a directed acyclic graph (DAG) that encodes the assumed relationships between contextual variables, role performance constructs, observed in-game statistics and outcomes. Due to space constraints, we provide the DAG in supplementary material. Contextual variables include game length, side advantage, pre-match win probability, team rating before the match and opponent rating before the match. These variables may affect both role performance and match outcome and therefore act as potential confounders. Role-performance affects its respective role-specific indicators, which in turn affect team-level quantities such as total kills and gold difference at 15 minutes. Match result affects post-match rating and therefore the rating gain. Let P_{ADC} and P_{JNG} denote the specific latent constructs, where each consists of a set of role specific indicators. P_{JNG} consists of `kills`, `assists`, `damage dealt per minute (DPM)` and `creep score per minute (CSPM)`, where creep score describes farming efficiency. P_{ADC} consists of the same indicators, adding `deaths`, `damage taken per minute`, `gold earned`.

We identify causal effects using backdoor adjustment [14], for which each set of treatment-outcome pair needs a sufficient adjustment set. The treatment variables are represented as $T_r, r \in \{adc, jng\}$, where each treatment variable represents the respective ADC or JNG performance construct. The outcomes are $Y_y, y \in \{win, gain\}$, where Y_{win} is the binary match outcome and Y_{gain} the rating gain. Additionally, we denote Z_i as the i-th adjustment set, where $i \in \{1, 2, 3, 4\}$ corresponds to the i-th causal question. An adjustment set is sufficient, if it blocks all non-causal backdoor paths from treatment to outcome. Non-causal backdoor paths are defined as all paths between T_r and Y_y that begin with an arrow into T_r and are not part of the causal effect of T_r on Y_y. All resulting sufficient adjustment sets contain `side_adv` and `win_prob`. When the outcome is Y_{win}, they also contain T_{JNG}, resp. T_{ADC}, and when the outcome is Y_{gain} they contain `rating_before` and `opp_rating_before` (see Tab 4 in the supplementary material).

The causal effect τ_i of the i-th causal question is estimated as the average treatment effect. We use the generalized definition of Hartwig [7] for continuous treatment variables. For each role $r \in \{\textsc{Adc}, \textsc{Jng}\}$ and outcome $y \in \{\text{win}, \text{gain}\}$, we define the target causal effect τ_i as the average marginal change in the interventional outcome expectation,

$$\tau_i = \frac{\partial}{\partial t_r} \mathbb{E}[Y_y \mid \text{do}(T_r = t_r)] \quad . \tag{1}$$

Using the backdoor adjustment, we can identify the effect of treatment T_r on outcome Y_y given a sufficient adjustment set Z_i, as follows:

$$\mathbb{E}[Y_y \mid \text{do}(T_r = t_r)] = \mathbb{E}_{Z_i} \left[\mathbb{E}[Y_y \mid T_r = t_r, Z_i] \right].$$

Applying the identified formula to Equation 1, the corresponding marginal causal effect is defined as:

$$\tau_i = \mathbb{E}_{Z_i} \left[\frac{\partial}{\partial t_r} \mathbb{E}[Y_y \mid T_r = t_r, Z_i] \right]. \tag{2}$$

We estimate the inner conditional expectation $\mathbb{E}[Y_y \mid T_r = t_r, Z_i]$ using regression models $Y_y \sim T_r + Z_i$, and average the resulting marginal effects over the empirical distribution of the adjustment variables. For the binary match outcome Y_{win} we use logistic regression, and for the continuous rating-gain outcome Y_{gain} we use linear regression. Estimating this model yields a fitted regression function $\hat{f}_i$, such that the function approximates the inner conditional expectation $\hat{f}_i(t_r, Z_i) \approx \mathbb{E}[Y_y \mid T_t = t_t, Z_i]$. We then compute the marginal effects for each observation and average them over the empirical distribution of the adjustment variables. This yields the following empirical estimator.

$$\hat{\tau}_i = \frac{1}{n} \sum_{k=1}^{n} \frac{\partial}{\partial t} \hat{f}_i(t, z_i^{(k)}) \quad , \tag{3}$$

where $z_i^{(k)}$ is the observed adjustment vector for sample k corresponding to adjustment set i, and n denotes the total number of observations. This is implemented using the DoWhy causal inference library [2, 17].

Our results depend on the correctness of the specified causal model based on domain knowledge and our causal assumptions. A particular limitation here is that the assumption of unconfoundedness cannot be tested. Instead, we can only verify that the results are statistically robust. To that end, we apply refutation tests which deliberately alter the data or model structure to check whether the estimated effects behave as expected under such changes. That is, we apply invariant transformations, i.e., Random Common Cause (RCC) and Data Subsample, which are expected to **not** influence the estimate, and nullifying transformations, i.e., Placebo Treatment, which should result in the estimate becoming 0.0 (see Sec. C in supplementary material).

We furthermore analyse the sensitivity to unobserved confounding, following the methods of Cinelli and Hazlet [5, 6]. This method specifically addresses the concern of unconfoundedness untestability by empirical means. It estimates how large an unobserved confounder would have to be to overturn the observed partial R^2 measures. That is, it estimates the minimum strength of confounding necessary to fully explain away the estimated effect.

Table 1. [Higher is Better] Latent role-performance constructs. CR denotes composite reliability and AVE denotes average variance extracted. Both constructs exceed the recommended CR and AVE thresholds reported by Cheung et al. [4]

Construct	AVE	CR
P_{ADC}	0.7025	0.9114
P_{JNG}	0.8524	0.9518

5 Empirical Results

We now present the main findings of this study, including the evaluation of the latent performance constructs, estimated causal effects, and robustness analyses.

We begin by assessing the quality of the latent Adc and Jng performance constructs using Construct Reliability (CR) and Average Variance Extracted (AVE). CR measures the internal consistency of a construct and indicates how reliably the observed indicators capture the same underlying latent variable. AVE measures the proportion of indicator variance explained by the latent construct and therefore reflects the degree to which the indicators represent the intended concept. We report the corresponding CR and AVE values for each construct in Tab. 1. We also examined the factor loadings during model development and found no evidence of problematic indicators. Both constructs comfortably exceed the recommended CR and AVE thresholds reported by Cheung et al. [4]. These results suggest that the latent performance constructs provide a sound foundation for the subsequent causal analysis.

The main results of this work are the estimated causal effects for each of the four causal questions, which we summarize in Table 2. Since the latent Adc and Jng performance variables are standardized, each estimate corresponds to the effect of a one-standard-deviation increase in the respective role-performance construct. For the binary match outcome, the estimates represent the average marginal effects on win probability, while for the rating gain Δ_{Rating} the estimates show the change in the standardized rating-gain variable. As a sanity check, we see that, the estimated effects on match victory are positive for both roles. Adc performance has the largest estimated effect on match victory: increasing the Adc performance score by one standard deviation increases the probability of winning by approximately 27.53 percentage points. For Jng performance the estimated effect on match victory is positive, but the increase of approximately 7.32 percentage points is comparatively smaller than that of Adc.

The estimated effects on rating gain are also positive, but substantially smaller. A one standard-deviation increase in Adc performance changes standardized rating gain by only 0.0033, while the corresponding estimate for Jungle performance is 0.0041. These values are numerically close to zero, underlining a contrast of applicability between the two measures of in-game role impact.

Table 2. Estimated causal effects of latent Adc and Jungle performance. Increasing the Adc performance score by one standard deviation increases the probability of winning by approximately 27.53%. For Jng performance the estimated effect is 7.32%.

Question	Treatment	Outcome	ATE
Q1	Adc performance	Victory	0.2753
Q2	Jng performance	Victory	0.0732
Q3	Adc performance	Δ_{Rating}	0.0033
Q4	Jng performance	Δ_{Rating}	0.0041

Lastly, we highlight the findings of the robustness tests, both for the refutation and the sensitivity testing. Table 3 showcases the results of the refutation testing, where we applied the Placebo Treatment Refuter, the Random Common Cause Refuter and the Data Subsample Refuter. The refuters are described in more detail in supplementary material. The table outlines the original estimate, the estimate under refutation for the Placebo Treatment Refuter, as well as the relative deviation of the estimate under refutation to the original estimate for both the Random Common Cause Refuter and the Data Subsample Refuter. All refuters produce results inline with the expected behaviour, supporting the stability of the estimated effects.

Table 3 reports three measures of sensitivity to unobserved confounding, following the guidelines of Cinelli and Hazlett [6]. The measure $R^2_{Y\sim T|X}$ describes how much of the remaining variance in the outcome Y is explained by treatment T, after accounting for confounders X. The remaining measures (RV and $RV_{\alpha=0.05}$) are as defined in Section 4. The results show a clear distinction between the effects on match victory, first and second causal question, and the effects on rating gain, third and fourth causal question. The estimated effect of ADC performance on match victory is shown to be highly robust, with a confounder needing to explain approximately 33.7% of the residual variance, in both treatment and outcome, to reduce the effect to zero. On the other hand, the effect of Jng performance on match victory is moderately stable, requiring an unobserved confounder explaining approximately 13.9% of the residual variance. In contrast, the effects on rating gain are shown to be extremely fragile to unobserved confounding, highlighting the mentioned distinction. The Adc performance effect on rating gain can be explained away by an unobserved confounder accounting for only 0.4% of the residual variance, and the effect of Jng performance by one only accounting for 0.6%. With the addition of the $R^2_{Y\sim T|X}$ values being 0.0, the estimated effects for rating gain are treated as directionally consistent, but not substantively reliable.

6 Discussion and Conclusion

Our findings do not support the broad claim that Adc is inherently weak in professional LoL. Under the available measurement model, strong ADC perfor-

Table 3. Refutation(Left, lower is better) and **Sensitivity** (Right, higher is better) analysis for each causal question. σ denotes the relative standard deviation and RCC denotes the Random Common Cause refuter. We see that estimated effects remain stable under all refutation tests. For sensitivity analysis, RV denotes the minimum confounder strength required to explain away the estimated effect, with larger values indicating greater robustness. $R^2_{Y\sim T|X}$ measures the adjusted treatment–outcome association. Although not a sensitivity metric, larger $R^2_{Y\sim T|X}$ generally correspond to effects that are harder to overturn.

	Refutation Tests ↓				Sensitivity ↑		
Question	Estimate	Placebo	σ_{RCC}	$\sigma_{\text{Subsample}}$	$R^2_{Y\sim T\mid X}$	RV	$RV_{\alpha=0.05}$
Q1	0.2753	0.00	1.2×10^{-6}	0.0003	0.147	0.337	0.334
Q2	0.0732	0.00	3.0×10^{-6}	0.0002	0.022	0.139	0.134
Q3	0.0033	≈ 0	1.8×10^{-5}	0.0100	0.000	0.004	0.000
Q4	0.0041	≈ 0	1.11×10^{-4}	0.0029	0.000	0.006	0.001

mance has a large and robust estimated direct effect on winning. Jungle performance also contributes positively, but its measured direct effect is smaller and less robust. A plausible interpretation is that the ADC role is more directly reflected in outcome-proximal statistics such as damage, kills, gold, and farming efficiency, whereas Jungle impact may be more indirect, enabling teammates through map pressure, ganks, tempo, and objective setup.

This interpretation is important because the result should not be read as a definitive ranking of intrinsic role importance. Our estimation of Jungle performance is limited by the available data: objective control, neutral-objective timing, map pressure, and other macro-level events are not fully represented across seasons. Since these are central responsibilities of the Jungle role, the estimated Jungle effect likely captures only the measurable direct component of Jungle performance. The analysis rather supports the more precise conclusion that measured isolated ADC performance has a larger and more robust direct causal effect on match victory than estimated Jungle performance.

Overall, the results demonstrate that combining latent variable modelling with graphical causal inference can produce interpretable estimates of role impact in professional E-sports. The approach moves beyond predictive associations and makes the required assumptions explicit. Future work should incorporate richer event-level data, patch- and champion-specific controls, draft information, and more complete objective-control variables to better capture indirect and strategic forms of role impact, especially for Jungle performance.

Bibliography

[1] Bahrololloomi, F., Klonowski, F., Sauer, S., Horst, R., Dörner, R.: E-sports player performance metrics for predicting the outcome of league of legends matches considering player roles. SN Computer Science 4(3), 238 (2023)

[2] Blöbaum, P., Götz, P., Budhathoki, K., Mastakouri, A.A., Janzing, D.: Dowhy-gcm: An extension of dowhy for causal inference in graphical causal models. Journal of Machine Learning Research 25(147), 1–7 (2024)

[3] Briel, P.: Your guide to playing every role in league of legends (October 2025), https://www.redbull.com/int-en/league-of-legends-roles-explained, red Bull esports guide explaining the five roles in *League of Legends*

[4] Cheung, G.W., Cooper-Thomas, H.D., Lau, R.S., Wang, L.C.: Reporting reliability, convergent and discriminant validity with structural equation modeling: A review and best-practice recommendations. Asia Pacific Journal of Management 41(2), 745–783 (2024)

[5] Cinelli, C., Ferwerda, J., Hazlett, C.: sensemakr: Sensitivity Analysis Tools for Regression Models (2026), r package version 0.1.6

[6] Cinelli, C., Hazlett, C.: Making sense of sensitivity: Extending omitted variable bias. Journal of the Royal Statistical Society: Series B (Statistical Methodology) 82(1), 39–67 (2020)

[7] Hartwig, F.P.: A generalized definition of the average causal effect for both binary and continuous treatments (2021)

[8] Herbrich, R., Minka, T., Graepel, T.: Trueskill™: A bayesian skill rating system. In: Advances in Neural Information Processing Systems 19. pp. 569–576 (2006)

[9] Insider, E.: League of legends roles explained & who does them best (2025), https://esportsinsider.com/league-of-legends-roles-explained

[10] Joshy, V.: Openskill: A faster asymmetric multi-team, multiplayer rating system. Journal of Open Source Software 9(93), 5901 (2024)

[11] Mobalytics: How to choose your main role for ranked league of legends (January 2026), https://mobalytics.gg/lol/guides/roles, online guide explaining the five primary roles (Top, Jungle, Mid, ADC, Support) and how to understand them

[12] Novak, A.R., Bennett, K.J.M., Pluss, M.A., Fransen, J.: Performance analysis in esports: Modelling performance at the 2018 league of legends world championship. International Journal of Sports Science & Coaching 15(5–6), 809–817 (2020)

[13] Oracle's Elixir: Oracle's elixir: Professional league of legends statistics. https://oracleselixir.com/ (2025), accessed for professional League of Legends match data

[14] Pearl, J.: Causality: Models, Reasoning, and Inference. Cambridge University Press, Cambridge, 2 edn. (2009)

[15] Reitman, J.G., Anderson-Coto, M.J., Wu, M., Lee, J.S., Steinkuehler, C.: Esports research: A literature review. Games and Culture 15(1), 32–50 (2020)

[16] Riot Games, Inc.: League of legends. https://www.riotgames.com (2009–present), original game publisher

[17] Sharma, A., Kiciman, E.: Dowhy: An end-to-end library for causal inference. In: Causal Data Science Meeting (2020), conference presentation

[18] Sharpe, B.T., Nicholson, M., Leis, O., Poulus, D., Birch, P.D.J., McNulty, C.: Indexing league of legends performance: A systematic review and focus group study. International Journal of Sports Science & Coaching 21(1) (2025)

A League of Legends Gameplay

This section introduces the core game mechanics and player roles, necessary for the analysis in this thesis. The focus is on match structure, key objectives, roles, and the professional competitive context.

A.1 Match Structure & Objective

The standard game mode examined in this thesis is played by two teams, each consisting of five players. Each player selects an individual champion, all having four abilities. League of Legends has currently more than 170 available unique champions, each with a distinct playstyle, item-builds and strengths. This creates a large variety of potential team compositions and interactions with opposing teams. Especially in professional play the composition of champions is important, with every professional draft having an essential identity and playstyle.

The map, as seen in Figure 1, is divided into two sides (red and blue) separated by a river, with three lanes, Top, Mid, and Bot, connecting the teams bases. Each lane contains three turrets and one inhibitor per side. In Figure 1, the turrets are denoted by rectangles and inhibitors by circles, where the color of the structure corresponds to the team it belongs to. Each lane is progressed by destroying the structures in a lane, where destruction grants gold directly, and pressure indirectly. The structures can only be destroyed in sequential order, from furthest to closest to the base of the team the structure belongs to, meaning the inhibitor can only be destroyed after the three turrets of the team, on that specific lane, have already been destroyed. Turrets deal significant damage to enemy units, providing defensive zones for teams. As seen in Figure 1, the lanes are labeled "Top Lane", "Mid Lane" and "Bot Lane", which is short for Bottom Lane.

The four separate areas in between the lanes, marked by the dotted lines in Figure 1, are called the Jungle area, where the color again denotes the corresponding team. The jungle contains neutral monsters that spawn at regular intervals, with epic monsters/neutral objectives spawning at fixed times in the river pits and providing team-wide advantages when defeated. Lastly minions spawn in each lane periodically throughout the game.

The term "farming" refers to killing the minions or normal monster, depending on the role, and the term "farm" refers to the amount of monsters or minions killed, being measured by the creep score. Both monsters and minions grant gold and experience when slain, representing the most consistent form of gold generation.

Gold enables item purchases, which in turn increase champion power. Gold can be obtained by farming either minions or monsters, as well as by destroying turrets. Additionally, killing opposing players, as well as assisting in the kill, grants gold. The amount of gold for a kill varies depending on game state and specific player performance, but the kill will always grant more gold then the assist.

Another concept essential to League of Legends is vision control, shown generally through the aggregated metric of visionscore. Vision control describes the amount of vision a team has on the map, as well as the amount of vision destroyed of the opposing team. To obtain additional vision, different types of "wards" can be used, which differ in amount of vision granted and time for which vision is granted. These wards can be destroyed through multiple interactions, including items, runes and other game mechanics.

The game is generally defined into three phases, commonly agreed upon in the community:

1. **Early Game / Laning Phase ($\sim 0 - 15$ min.):** Focus on farming and lane control by generating advantages, with occasional minor skirmishes
2. **Mid Game / End of Laning Phase ($\sim 15 - 20$ min.):** Focus around contesting neutral objectives, team rotations and the resulting team fights
3. **Late Game ($\sim 20+$ min.):** Fights and neutral objectives become critical, and have the ability to decide the game outcome, with most fights being team fights.

Lastly, the ultimate goal of a match is to destroy the enemy Nexus, located deep in each teams base and denoted by a respectively colored hexagon in Figure 1. Nexus destruction requires first eliminating at least one inhibitor, as well as destruction of the two turrets defending the Nexus.

A.2 Overview of the Roles and Team Compositions

In this section we will build on the foundation laid by the previous section, by giving an overview of the specific roles and the basics of team compositions. As explained in Section A.1, as well as seen on Figure 1, the map is divided into four general areas per side, the three lanes and the jungle area. In respect to these areas, the following five roles have been established

1. Top(-Laner)
2. Mid(-Laner)
3. Bot/Bottom(-Laner)
4. Support
5. Jungler

The role names generally indicate the primary map position, in which a player operates during the early stages of the game. For example, the Jungler mainly resides in the jungle area and the Mid Laner in the mid lane. While each role is associated with typical responsibilities, there usually is no single fixed champion archetype assigned to a role, as champion selection depends on team strategy, synergy considerations, and tactical objectives. The following will give a short explanation and overview of all of the five roles, as agreed upon in the community[9, 11, 3]:

- **Top(-Laner):** Usual champions are characterized by durability, and are good at operating in isolation. Contributing through individual map pressure

or frontline presence in team fights, depending on team composition and opponent.
- **Jungler:** Operates between lanes and gains resources through periodically farming neutral camps. Characterized by map pressure on all lanes and the responsibility of contesting neutral objectives. Champion type is diverse, but typically provides some aspects of mobility, damage output and crowd control utility.
- **Mid(-Laner):** Plays in the central lane and typically uses champions providing some aspects of play-making ability, team fighting utility or map mobility. Assisting in other lanes and neutral objectives, due to central position.
- **Bot(-Laner) / Attack Damage Carry (ADC):** Typically Marksman-type champions having high amounts sustained ranged damage output, with low defensive stats. Paired with support role in the Bot-Lane, playing key role in team fights and scaling with gold and items.
- **Support:** Assists the ADC in the early game, with the ability to assist other lanes as needed. Provides vision, protection utility and crowd control depending on the specific champion played.

Team compositions are primarily shaped by a team's strategic objectives and intended win conditions. Champions are selected to create synergistic interactions and to support specific tactical approaches, rather than to fulfill rigid archetypal templates. Many compositions revolve around particular win conditions, such as prioritizing utility for a strong central champion, focusing on coordinated team fighting, or emphasizing early or late-game advantages.

A.3 Context of Professional Competition

This study exclusively uses professional League of Legends match data, rather than ranked or casual match data. The primary reason for this choice, is data quality and behavioral consistency. Professional matches are played in a highly controlled competitive environment, in which all participants are incentivized to win and perform optimally, in terms of their role and responsibilities. We believe that non-professional matches have substantially higher noise, due to non-competitive behavior, unstructured team fighting or other factors.

Professional play also ensures a consistently high mechanical and strategic skill level across all roles. Thus, observed performance differences are more likely to reflect meaningful strategic and executional variation, rather than basic skill gaps or misunderstanding of role responsibilities. This makes role-based impact comparisons, such as between ADC and Jungle, more valid and interpretable.

Another important factor is draft quality and role clarity. Professional teams use structured drafting processes, supported by coaching staff, scouting, and preparation. Champion selections are therefore intentional and strategically coherent, rather than preference driven or random. This reduces uncontrolled variance from poor compositions and increases the likelihood, that each role is placed in a composition designed to function properly, which strengthens internal validity when attributing outcome effects to specific roles.

Additionally, objective control, vision setup, lane assignments, and resource allocation are executed according to established strategic principles, which creates more stable and comparable game states across matches, making it easier to model how ADC and Jungle contributions translate into win probability. In lower tier play, chaotic decision making and inconsistent macro behavior introduce confounding variation, which weakens causal inference.

Lastly, the professional context helps satisfy a key assumption for causal analysis: that observed behavior reflects genuine competitive intent rather than arbitrary or low effort play.

For these reasons, higher behavioral reliability, strategic structure, draft quality, role clarity, statistical accuracy, and incentive alignment, professional League of Legends matches provide a suitable dataset for estimating the causal impact of ADC and Jungle roles on match outcomes.

A.4 Role Specific Performance Definition

As this thesis aims to investigate the impact of player roles on match outcomes, with a particular focus on the Jungle and ADC roles, it is necessary to define what constitutes role-specific impact. Sharpe et al. [18] discusses that in League of Legends, player performance cannot be fully captured by single quantitative metrics alone, as the effectiveness of a player depends on how well they execute their responsibilities inherent to their assigned role. Therefore, this section examines the core responsibilities, decision making priorities, and strategic functions of the ADC and Jungle roles. By understanding these roles in the context of their typical in-game duties, we can establish a framework for evaluating player impact, based on the successful fulfillment of the responsibilities, rather than relying solely on raw statistics such as kills or gold earned.

The specification of role responsibilities is based on domain knowledge as well as community blogs or post [9, 11, 3], as there is no concrete definition by Riot Games.

The ADC or the Bot-Lane role is typically occupied by a marksman champion, as mentioned in Section A.2. The marksman champion type is characterized by dealing high and consistent amounts of damage from range, with low defensive stats. Additionally, guides and blogs similar to [9, 3] outline the characteristic of scaling with gold and items and thus the responsibility of farming efficiently to faster achieve the power accompanied with the items. As explained before, starting in the mid-game team fights become increasingly more important, as does the presence of the ADC in team fights. The ADC role is expected to be the main consistent damage source, which results in the ADC being the highest priority target for opposing teams, as the longer an ADC is alive the more damage they could potentially deal. Thus, the last crucial aspect of the ADC role is positioning. Conclusively, we identify the responsibilities of the ADC role as follows:

- Efficient farming and gold acquisition
- Providing sustained ranged damage output, especially in team fights

– Maintaining safe positioning to maximize damage up-time and survivability

Continuing with the Jungle role, there exists a wide range of viable champions and no single dominant archetype is universally preferred. The Jungler differs from the lane roles in not occupying a fixed lane, but instead operating primarily in the jungle area between lanes. This structural difference provides greater movement flexibility across the map.

This flexibility is reflected in the Jungler's responsibilities, which include creating advantages for teammates through coordinated lane pressure, commonly referred to as ganking. A central aspect of the role, is neutral objective control, including monsters that provide team wide benefits, as also emphasized in role descriptions [9, 11].

As the role must balance resource collection with map impact, Junglers are required to make continuous decisions between farming neutral camps, applying pressure to lanes, and securing objectives. Efficient Jungle pathing and timing of map movements are therefore considered core skills associated with the role. Based on these commonly described characteristics, the Jungle role responsibilities can be summarized as:

– Applying map pressure through ganks and lane assistance
– Securing and contesting neutral objectives
– Managing jungle farm efficiently through pathing and timing decisions

The defined responsibilities of both roles guide the extraction of role-specific metrics and evaluation of player impact in the later analysis.

B Causal Graph

<Osman: I would suggest writing down full forms of variables in this section as well, in case a reader is not familiar with acronyms. A simple list should do.>

C Details on Refutation Tests

Table 4. Causal questions and sufficient adjustment sets.

Effect	Treatment	Outcome	Adjustment set
τ_1	T_{ADC}	Y_{win}	$\{\texttt{side\_adv}, \texttt{win\_prob}, T_{\text{JNG}}\}$
τ_2	T_{JNG}	Y_{win}	$\{\texttt{side\_adv}, \texttt{win\_prob}, T_{\text{ADC}}\}$
τ_3	T_{ADC}	Y_{gain}	$\{\texttt{side\_adv}, \texttt{win\_prob}, \texttt{rating\_before}, \texttt{opp\_rating\_before}\}$
τ_4	T_{JNG}	Y_{gain}	$\{\texttt{side\_adv}, \texttt{win\_prob}, \texttt{rating\_before}, \texttt{opp\_rating\_before}\}$

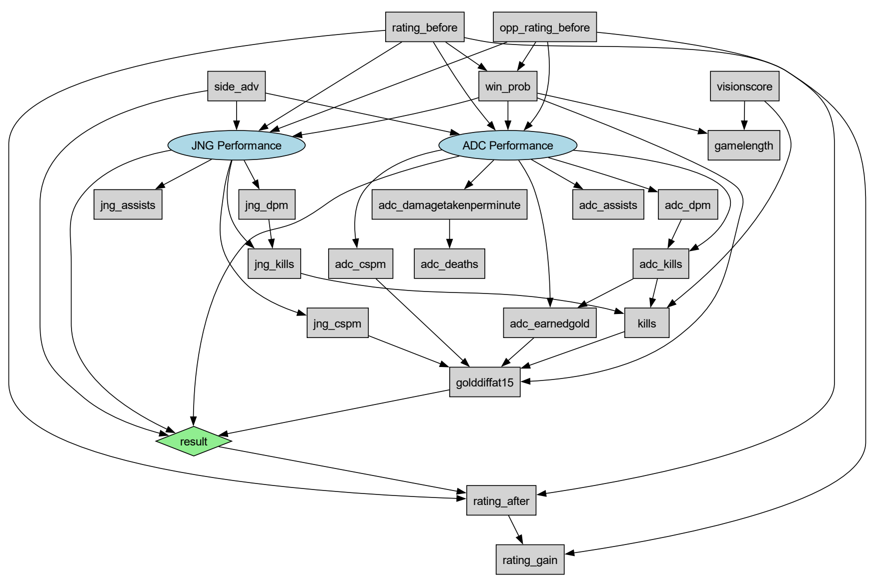

Fig. 2. Full causal model including ADC and JNG latent performance variables. Ellipses denote latent variables, boxes denote observed indicators. A directed edge $X \to Y$ indicates that X is a cause of Y.

1. **Placebo Treatment:** Replaces the treatment with a random variable. The estimated effect should collapse to approximately zero, indicating that the original effect is linked to the treatment rather than statistical artefacts.
2. **Random Common Cause:** Introduces a random variable as an additional confounder. The estimated effect should remain stable, indicating that the estimate is robust to irrelevant covariates.
3. **Data Subsample:** Re-estimates the effect on random subsets of the dataset. The estimated effect should remain consistent across subsamples, indicating that the result is not driven by a subset of matches.

D Details on Sensitivity Analysis

The methods of Cinelli and Hazlet [6] propose analysing partial R^2 measures for the results, estimating how large an unobserved confounder would have to be to overturn the results. The method assumes for an unobserved confounder U to exist. We then calculate the following measures:

1. **Treatment Association:** $R^2_{T_r \sim U|X}$ = proportion of the residual variance in the treatment T_r explained by U after controlling for X
2. **Outcome Association:** $R^2_{Y_y \sim U|T_r,X}$ = proportion of the residual variance in the outcome Y explained by U after adjusting for T and X

where X is the set of observed confounders. Using these measures a robustness value (RV_q) is defined, which describes how strong the confounder's partial R^2 would have to be, equal for both treatment and outcome association, to reduce the effect by a proportion q. In simpler terms, $RV_{q=1}$ is the minimum strength of confounding necessary to fully explain away the estimated effect. In this work, we only consider $RV_{q=1}$, which is referred to as simply RV, and $RV_{\alpha=0.05}$, which describes the strength necessary to make the result statistically insignificant for $\alpha = 0.05$.